

Session Objectives

1. Describe the epidemiology associated with rheumatic fever, rocky mountain spotted fever, and Chagas disease.
2. Identify and describe the microbial pathogens associated with each disease.
3. Describe the pathogenesis and clinical presentation associated with each disease.
4. Explain the diagnosis and treatment for each disease.

Acute Rheumatic Fever (ARF)

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JOINT PAIN + FEVER CAN BE:
RHEUMATIC FEVER

Malaria is not the only cause of joint pain and fever. Joint pain and fever can indicate **Acute Rheumatic Fever**.

Two atipa pee n keken aye kelo arem me wang mwot kede lyeto ento twero bedo anyut me **two adunu nyo two itao.**

Acute Rheumatic Fever can damage the heart.

Two adunu nyo two itao twero balo adunu ni woko.

If your child (3-17 yrs) has joint pain and fever take them to the nearest health center for evaluation.

Cwal atini (me mwaka 3 nio 18) matye kede arem me wang mwot kede lyeto I ot yat ma cok kedi wek opim.

FREE EVALUATIONS IN LIRA REGIONAL REFERRAL HOSPITAL

Dakatal adwong me Lira
TYE ME NONO

STaP Rheumatic Heart Disease
Prevention • Treatment • Research

0800-201900

Logos of various Nigerian health organizations and the Global Network.

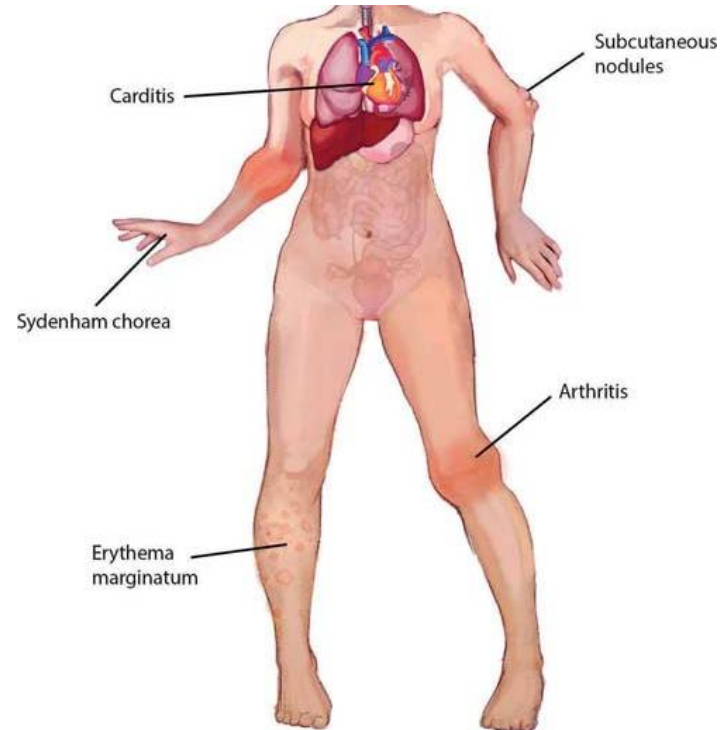
- Multisystem disease
- Results from an autoimmune-like reaction to *Streptococcus pyogenes* infection
- Occurs around **3 weeks** after group A *Streptococcus pyogenes* pharyngitis
- Mainly a disease of **children** aged 5-15 y/o
- Higher incidence in low-resource countries

Longo, D., Fauci, A., Kasper, D., Hauser, S., Larry, J. J., Loscalzo, J., Holland, S., & Langford, C. (2025).

Harrison's Principles of Internal Medicine, 22nd Edition.

ARF: JONES criteria

- **JONES Criteria:** rules for diagnosis of ARF
 - Required criteria and 2 major or
 - Required criteria, 1 major and 2 minor
- Required: evidence of prior group A streptococcus infection
- Major: J ❤️ NES
- Minor:
 - Fever
 - Arthralgia
 - Previous RF/RHD
 - Acute phase reactants (ESR, CRP)
 - Prolonged PR interval



RMSF Clinical Presentation

- Initially, similar to a self-limiting viral illness: fever, headache, malaise, etc.
- Rash** (macules) appears first on the **wrists and ankles**, then spreads to the remainder of the extremities and trunk
 - Rash is due to **vasculitis** and inflammatory damage to the blood vessel
 - Can become necrotic vasculitis -> CNS, liver and renal damage

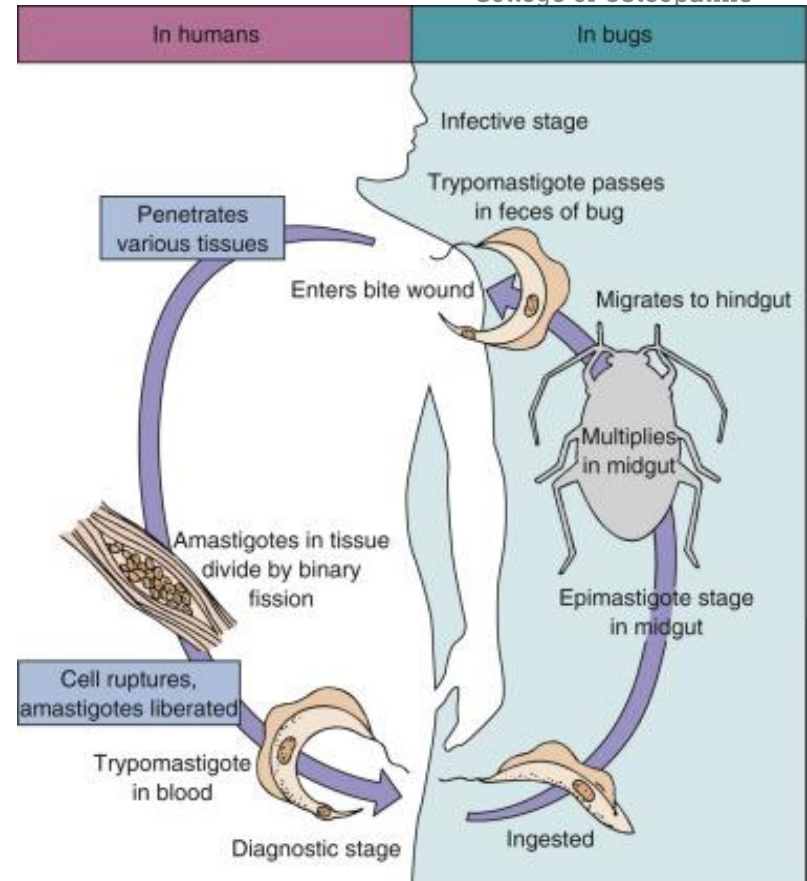


Chagas Disease: Pathophysiology

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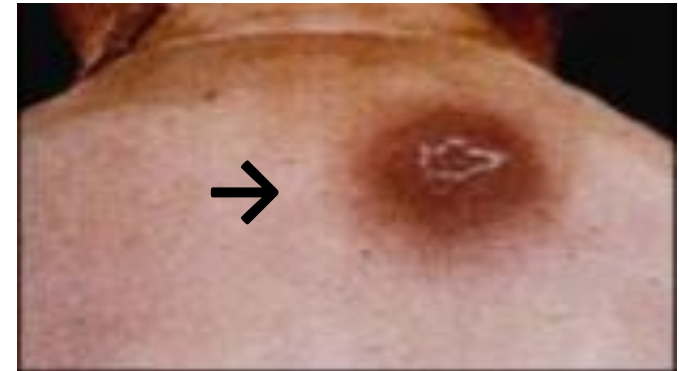
- Vector = **reduviid bug** (AKA kissing bug or assassin bug)
 - Bites around the mouth and on the face
 - Trypomastigotes in bug's feces
- Protozoa enters bloodstream/lymphatics
 - Cardiac muscle, liver, brain
 - Forms "cysts"



Chagas Disease: Clinical presentation

(Acute)

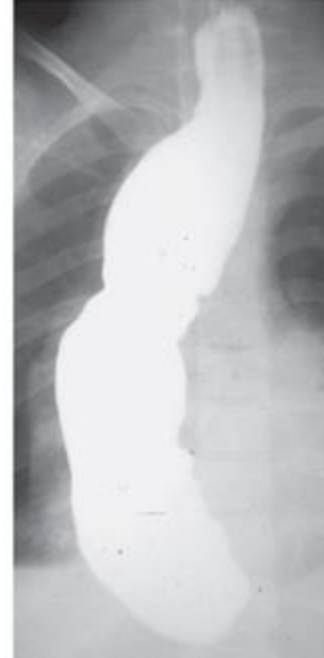
- Romaña sign: unilateral painless edema of periocular tissue (→)
- Chagoma: indurated area with local lymphadenopathy at the site of the bite (→)
- Malaise, fever, anorexia, generalized lymphadenopathy, hepatosplenomegaly
- Acute signs and symptoms usually resolve spontaneously until development into chronic chagas



Chagas Disease: Clinical Presentation

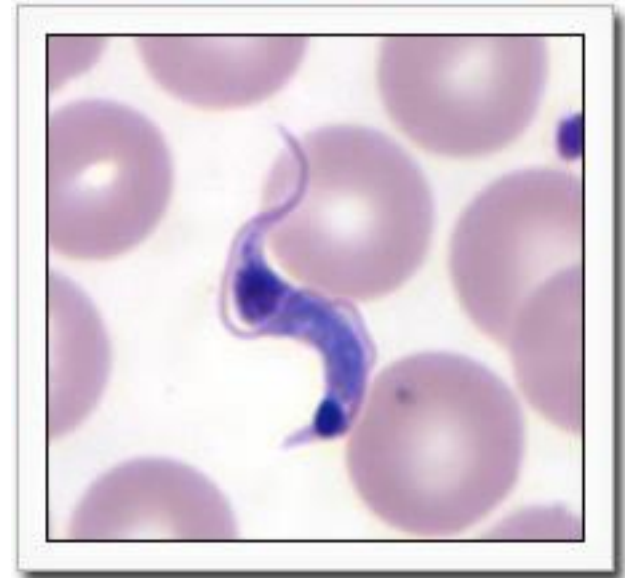
(Chronic)

- Years to decades after initial infection
- Cardiac arrhythmias
- Myocarditis
- Dilated cardiomyopathy
- Megaesophagus
- Megacolon



Chagas Disease: Diagnosis and Treatment

- Diagnosis
 - Visualize protozoa in blood, lymph node, liver, spleen, bone marrow using the Giemsa stain
 - Serological
 - PCR
- Treatment
 - Nifurtimox or benznidazole for acute cases



Summary Slide

Disease	Organism	Clinical Features	Diagnosis	Treatment
Rheumatic Fever	<i>Streptococcus pyogenes</i>	Migratory polyarthritits, carditis, subcutaneous nodules, erythema marginatum, Sydenham chorea	JONES criteria; prior strep infection	Penicillin
Rocky Mountain Spotted Fever	<i>Rickettsia rickettsii</i> Vector: <i>Dermacentor variabilis</i>	Rash that begins on the wrists and ankles then spreads to the rest of the extremities and trunk	Serology; tissue biopsy; PCR; prior tick exposure	Doxycycline
Chagas Disease	<i>Trypanosoma cruzi</i> Vector: Reduviid Bug	Acute: Chagoma, Romaña sign; Chronic: arrhythmias, Dilated cardiomyopathy, Megacolon, Megaesophagus	Visualize protozoan in blood/tissue; serology; PCR	Nifurtimox or Benznidazole